WoSM Academy

M201-F27 · Calculus II

Absolute and conditional convergence

Alternating series and rearrangements

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Lecture 03 · Unit 3 · Fall 2027

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Last time

For series with $a_k \geq 0$:

- $\sum a_k$ converges if and only if its partial sums are bounded.
- Integral test; $\sum \frac{1}{k^p}$ converges if and only if $p > 1$.
- Comparison and limit comparison with known series.
- Ratio and root tests: $L < 1$ converges, $L > 1$ diverges, $L = 1$ says nothing.

Every one of these needed $a_k \geq 0$. Today: series whose terms have both signs.

This lecture

01 Absolute convergence

02 Alternating series

03 Rearrangements

04 Summary and practice

Reading: G. Strang, E. Herman, *Calculus, Volume 2* (OpenStax), Section 5.5.

01

Absolute convergence

Terms of both signs

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots, \qquad \sum_{k=1}^{\infty} \frac{\sin k}{k^2}, \qquad \sum_{k=0}^{\infty} \frac{(-2)^k}{k!}$$

- The partial sums no longer increase, so "bounded partial sums" is not enough: $1 - 1 + 1 - \cdots$ has bounded partial sums and diverges.
- The tests of Lecture 2 need $a_k \geq 0$.

Idea: first look at the *sizes* of the terms, $|a_k|$. Cancellation between positive and negative terms can only help.

Absolute and conditional convergence

Definition

The series $\sum a_k$ **converges absolutely** if $\sum |a_k|$ converges. It **converges conditionally** if $\sum a_k$ converges but $\sum |a_k|$ diverges.

- If $a_k \geq 0$, convergence and absolute convergence are the same thing.
- $\sum |a_k|$ has nonnegative terms, so all the tests of Lecture 2 apply to it.

Theorem

If $\sum a_k$ converges absolutely, then it converges, and $\left|\sum_{k=1}^{\infty} a_k\right| \leq \sum_{k=1}^{\infty} |a_k|$.

Proof

Proof.

Let $\varepsilon > 0$. Since $\sum |a_k|$ converges, the Cauchy criterion (Lecture 1) gives an N with $|a_{m+1}| + \cdots + |a_n| < \varepsilon$ for $n > m \geq N$. By the triangle inequality, $|a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n| < \varepsilon$: $\sum a_k$ satisfies the Cauchy criterion too, so it converges. Finally $|S_n| \leq \sum_{k=1}^{n} |a_k| \leq \sum_{k=1}^{\infty} |a_k|$; let $n \to \infty$. □

A proof without Cauchy

$0 \leq a_k + |a_k| \leq 2|a_k|$, so $\sum (a_k + |a_k|)$ converges by comparison. Then $a_k = (a_k + |a_k|) - |a_k|$ is a difference of two convergent series.

Examples

- $\sum \dfrac{\sin k}{k^2}$: $\left|\dfrac{\sin k}{k^2}\right| \le \dfrac{1}{k^2}$, so it converges absolutely.
- $\sum \dfrac{(-1)^k}{2^k}$: $\sum \frac{1}{2^k}$ converges, so it converges absolutely (to $\frac{1}{1+1/2} = \frac{2}{3}$ from $k = 0$).
- $\sum \dfrac{\cos(k^2)}{k\sqrt{k}}$: $|a_k| \le \dfrac{1}{k^{3/2}}$, a p-series with $p > 1$. Converges absolutely.
- $\sum \dfrac{(-1)^{k+1}}{k}$: $\sum \frac{1}{k}$ diverges. The theorem says nothing — see Section 2.

Warning

"$\sum |a_k|$ diverges" does *not* mean that $\sum a_k$ diverges.

Ratio and root tests for any series

Theorem

Suppose $\left|\frac{a_{k+1}}{a_k}\right| \to L$ or $\sqrt[k]{|a_k|} \to L$. If $L < 1$, then $\sum a_k$ converges absolutely. If $L > 1$, then $\sum a_k$ diverges.

Proof: for $L < 1$, apply Lecture 2 to $\sum |a_k|$. For $L > 1$, Lecture 2 showed $|a_k| \not\to 0$, so $a_k \not\to 0$.

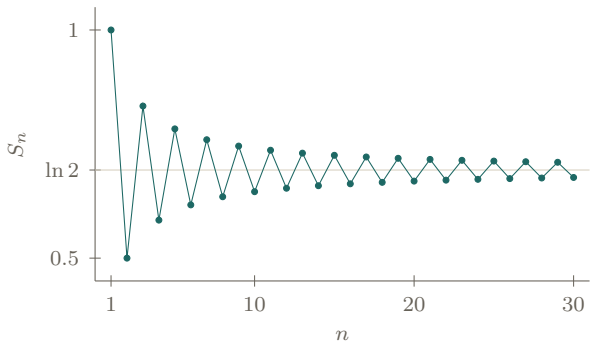
Example

- $\sum \frac{(-3)^k}{k!}$: $\left|\frac{a_{k+1}}{a_k}\right| = \frac{3}{k+1} \to 0$. Converges absolutely.
- $\sum \frac{(-2)^k}{k^2}$: $\left|\frac{a_{k+1}}{a_k}\right| = \frac{2k^2}{(k+1)^2} \to 2$. Diverges.

02

Alternating series

The alternating harmonic series



$$1 - \tfrac{1}{2} + \tfrac{1}{3} - \tfrac{1}{4} + \cdots$$

- The odd partial sums $S_1, S_3, S_5, \ldots$ decrease.
- The even ones $S_2, S_4, S_6, \ldots$ increase.
- Each step is shorter than the last, so they close in on one number.

The sum is $\ln 2 \approx 0.6931$ (Unit 6) — yet $\sum \frac{1}{k} = \infty$.

The alternating series test

Theorem (Alternating series test, Leibniz)

Let (b_k) be decreasing with $b_k \to 0$ (so $b_k \geq 0$). Then $\sum_{k=1}^{\infty}(-1)^{k+1}b_k = b_1 - b_2 + b_3 - \cdots$ converges. Its sum S lies between any two consecutive partial sums, and

$$|S - S_n| \leq b_{n+1}.$$

- The error of stopping after n terms is at most the first term left out, and has the same sign as that term.
- Only the tail matters: it is enough that (b_k) decreases from some k on.

Proof

Proof.

- $S_{2n+2} - S_{2n} = b_{2n+1} - b_{2n+2} \geq 0$: the even partial sums increase.
- $S_{2n+1} - S_{2n-1} = -b_{2n} + b_{2n+1} \leq 0$: the odd partial sums decrease.
- $S_2 \leq S_{2n} \leq S_{2n} + b_{2n+1} = S_{2n+1} \leq S_1$: both sequences are bounded.

By monotone convergence $S_{2n} \to S'$ and $S_{2n+1} \to S''$, and $S'' - S' = \lim b_{2n+1} = 0$. So $S_n \to S = S' = S''$, with $S_{2n} \leq S \leq S_{2m+1}$ for all n, m. Hence S lies between S_n and S_{n+1}, and $|S - S_n| \leq |S_{n+1} - S_n| = b_{n+1}$. □

Examples

- $\sum \dfrac{(-1)^{k+1}}{k}$ and $\sum \dfrac{(-1)^{k+1}}{\sqrt{k}}$
  $\frac{1}{k}$ and $\frac{1}{\sqrt{k}}$ decrease to 0: both converge. $\sum \frac{1}{k}$ and $\sum \frac{1}{\sqrt{k}}$ diverge, so both converge *conditionally.*
- $\sum \dfrac{(-1)^k k}{k^2+1}$
  $\left(\frac{x}{x^2+1}\right)' = \frac{1-x^2}{(x^2+1)^2} \leq 0$ for $x \geq 1$, and $\frac{k}{k^2+1} \to 0$: converges. $|a_k| \sim \frac{1}{k}$: conditionally.
- $\sum \dfrac{(-1)^k \ln k}{k}$
  $\left(\frac{\ln x}{x}\right)' = \frac{1-\ln x}{x^2} < 0$ for $x > e$: decreasing from $k = 3$ on, and $\frac{\ln k}{k} \to 0$. Converges conditionally.

How many terms do we need?

Example ($\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k!} = 1 - \frac{1}{e}$ **(Unit 6))**

The error after n terms is at most $\frac{1}{(n+1)!}$. As $10! = 3\,628\,800$, nine terms give $1 - \frac{1}{e}$ to within 10^{-6}.

Example ($1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}$ **(Unit 6))**

The error after n terms is at most $\frac{1}{2n+1}$. For an error below 10^{-3} we need about 500 terms; for 10^{-6}, about 500 000.

Both converge; how fast depends on how fast $b_k \to 0$. The test always tells us how far we can be from the sum.

Both conditions matter

$b_k \to 0$

$\sum (-1)^k \frac{k}{k+1}$ alternates, but its terms do not tend to 0. It diverges.

(b_k) decreasing

$1 - 1 + \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{9} + \frac{1}{4} - \frac{1}{16} + \cdots$ alternates, and $b_k \to 0$, but (b_k) is not decreasing. Here $S_{2n} = \sum_{j=1}^{n} (\frac{1}{j} - \frac{1}{j^2}) = H_n - \sum_{j=1}^{n} \frac{1}{j^2} \to \infty$, as $H_n \to \infty$ and $\sum \frac{1}{j^2}$ converges. It diverges.

A map

Series	$\sum \lvert a_k \rvert$	$\sum a_k$	**Verdict**
$\sum \frac{(-1)^k}{k^2}$	converges	converges	absolutely convergent
$\sum \frac{(-1)^k k}{2^k}$	converges	converges	absolutely convergent
$\sum \frac{(-1)^k}{k}$	diverges	converges	conditionally convergent
$\sum \frac{(-1)^k}{\sqrt{k}}$	diverges	converges	conditionally convergent
$\sum (-1)^k$	diverges	diverges	divergent

In general $\sum \frac{(-1)^k}{k^p}$ converges absolutely for $p > 1$, conditionally for $0 < p \leq 1$, and diverges for $p \leq 0$.

03

Rearrangements

Changing the order

A finite sum does not depend on the order of its terms. What about a series?

Definition (Rearrangement)

Let $\sigma\colon \{1,2,3,\dots\} \to \{1,2,3,\dots\}$ be a bijection. The series $\sum_{k=1}^{\infty} a_{\sigma(k)}$ is a **rearrangement** of $\sum_{k=1}^{\infty} a_k$: the same terms, each used exactly once, in another order.

Example

σ swaps $2j-1$ and $2j$: $\quad a_2 + a_1 + a_4 + a_3 + a_6 + a_5 + \cdots$

Same terms, half the sum

$S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots = \ln 2$. Now take one positive term, then two negative ones:

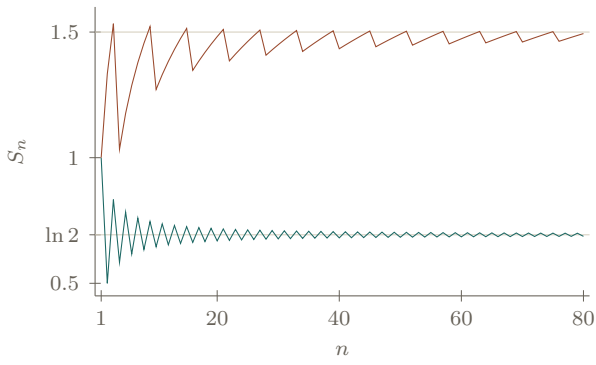
$$\Big(1 - \frac{1}{2} - \frac{1}{4}\Big) + \Big(\frac{1}{3} - \frac{1}{6} - \frac{1}{8}\Big) + \Big(\frac{1}{5} - \frac{1}{10} - \frac{1}{12}\Big) + \cdots$$

The j-th block is $\dfrac{1}{2j-1} - \dfrac{1}{4j-2} - \dfrac{1}{4j} = \dfrac{1}{4j-2} - \dfrac{1}{4j} = \dfrac{1}{2}\Big(\dfrac{1}{2j-1} - \dfrac{1}{2j}\Big)$, so the blocks add up to

$$\frac{1}{2}\Big(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\Big) = \frac{1}{2}\ln 2.$$

Inside a block the partial sums differ from the block sums by terms that tend to 0, so the rearranged series itself converges to $\frac{1}{2}\ln 2$.

Any sum you like



The same terms $1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \ldots$, in two orders:

- in the usual order: the sum is $\ln 2$;
- greedy: add positive terms $1, \frac{1}{3}, \frac{1}{5}, \ldots$ until the sum is above 1.5, then negative terms $-\frac{1}{2}, -\frac{1}{4}, \ldots$ until it is below, and repeat.

The greedy order converges to 1.5. Any target works.

Riemann's rearrangement theorem

Theorem (Riemann)

If $\sum a_k$ converges conditionally, then for every $c \in \mathbb{R}$ some rearrangement of it converges to c. Other rearrangements diverge to ∞ or to $-\infty$.

Idea of the proof. Split the nonzero terms into positive terms $p_1, p_2, \dots$ and negative terms $-q_1, -q_2, \dots$ $(q_j > 0)$, each in their original order.

- Both $\sum p_j$ and $\sum q_j$ diverge. Among the first n terms, if P_n and Q_n are the sums of the p's and q's, then $S_n = P_n - Q_n$. If (P_n) were bounded, so would $Q_n = P_n - S_n$ be, and then $\sum |a_k|$ would converge.
- So the greedy algorithm never gets stuck. Each time it crosses c it overshoots by at most the last term used, and the terms tend to 0.

Absolute convergence: order does not matter

Theorem

If $\sum a_k$ converges absolutely with sum S, then every rearrangement converges absolutely, also with sum S.

Proof.

Let $T_m = \sum_{k=1}^{m} a_{\sigma(k)}$ and $\varepsilon > 0$. Choose N with $\sum_{k>N} |a_k| < \varepsilon$, then M so large that $\sigma(1), \dots, \sigma(M)$ include $1, \dots, N$. For $m \geq M$, $T_m - S_N$ is a sum of terms a_k with $k > N$, so $|T_m - S_N| < \varepsilon$. Also $|S - S_N| < \varepsilon$, hence $|T_m - S| < 2\varepsilon$. The same argument for $\sum |a_{\sigma(k)}|$ gives absolute convergence. □

What this means in practice

- An **absolutely convergent** series behaves like a finite sum: you may reorder it, regroup it, or sum its positive and negative terms separately.
- In a **conditionally convergent** series the order is part of the series. Keep it.
- Power series (Units 5–7) converge absolutely inside their interval of convergence. That is why we can rearrange and multiply them freely.

> **On a computer**
>
> In floating-point arithmetic even a *finite* sum depends on the order, because each addition rounds. For a slowly converging series, adding the small terms first usually loses less accuracy. This is a different effect from Riemann's theorem.

04

Summary and practice

Summary

- $\sum a_k$ **converges absolutely** if $\sum |a_k|$ converges, and then it converges. Test $\sum |a_k|$ with the tests of Lecture 2.
- Ratio and root tests work for any series: $L < 1$ absolutely convergent, $L > 1$ divergent.
- **Alternating series test**: if b_k decreases to 0, then $\sum (-1)^{k+1} b_k$ converges, with error $|S - S_n| \le b_{n+1}$.
- **Conditionally convergent**: convergent, but not absolutely. Example: $\sum \frac{(-1)^{k+1}}{k} = \ln 2$.
- Rearranging an absolutely convergent series does not change its sum. A conditionally convergent series can be rearranged to any sum (Riemann).

Practice

Does the series converge absolutely, converge conditionally, or diverge?

1. $\displaystyle\sum \frac{(-1)^k}{k^{3/2}}$
2. $\displaystyle\sum_{k=2}^{\infty} \frac{(-1)^k}{\ln k}$
3. $\displaystyle\sum \frac{(-1)^k k}{k+5}$
4. $\displaystyle\sum \frac{\cos(k\pi)}{\sqrt{k}}$
5. $\displaystyle\sum \frac{(-5)^k}{k!}$
6. $\displaystyle\sum \frac{\sin k}{k^3}$
7. $\displaystyle\sum (-1)^k(\sqrt{k+1}-\sqrt{k})$

Practice, continued

8. How many terms of $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^3}$ guarantee an error below 10^{-4}?
9. For which p does $\sum \frac{(-1)^k}{k^p}$ converge absolutely? Conditionally?
10. Show that if $\sum a_k$ converges absolutely, then $\sum a_k^2$ converges. Is this still true if $\sum a_k$ only converges conditionally?
11. Describe a rearrangement of $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ that diverges to ∞.
12. (Challenge) Show that $1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \cdots$ (two positive terms, then one negative) has sum $\frac{3}{2}\ln 2$. Hint: $H_n = \ln n + \gamma + \varepsilon_n$ with $\varepsilon_n \to 0$.

Answers

1. Absolutely: $\sum \frac{1}{k^{3/2}}$ converges.
2. Conditionally: $\frac{1}{\ln k}$ decreases to 0; $\frac{1}{\ln k} \geq \frac{1}{k}$.
3. Diverges: the terms do not tend to 0.
4. Conditionally: $\cos(k\pi) = (-1)^k$.
5. Absolutely: ratio $\frac{5}{k+1} \to 0$.
6. Absolutely: $\left|\frac{\sin k}{k^3}\right| \leq \frac{1}{k^3}$.
7. Conditionally: $\sqrt{k+1} - \sqrt{k} = \frac{1}{\sqrt{k+1}+\sqrt{k}}$ decreases to 0, and its sum telescopes to ∞ (Lecture 1).
8. 21 terms: $\frac{1}{22^3} = \frac{1}{10\,648} < 10^{-4}$, but $21^3 = 9261$.

Answers, continued

9. Absolutely for $p > 1$; conditionally for $0 < p \le 1$ (diverges for $p \le 0$).
10. $|a_k| \to 0$, so $|a_k| \le 1$ and $a_k^2 \le |a_k|$ for $k \ge N$; compare. Not true for conditional convergence: $a_k = \frac{(-1)^k}{\sqrt{k}}$ gives $\sum a_k^2 = \sum \frac{1}{k}$.
11. Add positive terms until the sum exceeds 1, then one negative term; positive terms until it exceeds 2, then one negative term; and so on. Each stage ends because $\sum \frac{1}{2j-1} = \infty$.
12. After n blocks the partial sum is $\sum_{j=1}^{2n} \frac{1}{2j-1} - \frac{1}{2}H_n = H_{4n} - \frac{1}{2}H_{2n} - \frac{1}{2}H_n \to \ln 4 - \frac{1}{2}\ln 2 = \frac{3}{2}\ln 2$.

Next time

Lecture 04 — Sequences and series of functions (Unit 4)

- Pointwise and uniform convergence
- When the limit of continuous functions is continuous; integrating and differentiating limits
- Series of functions and the Weierstrass M-test

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.